Apache, often considered the world's most lethal attack helicopter, showed the faith reposed in the LCH.

The Prachand has a tandem configuration (one behind the other) for the pilot and co-pilot, who is also the Weapon System Operator (WSO) and a narrow fuselage, giving it its sleek appearance. It carries a 20mm twin barrel turret gun, 68mm rockets and air-to-air missile systems like ATGMs (anti-tank guided missiles). It also has a number of stealth features such as reduced radar and infra-red signatures and crashworthy landing gear. The LCH has a maximum speed of 275kmph (148kt), a combat radius of 500km and an operational ceiling of 16,000 to 18,000 feet (5,490 metres).

A key HAL official says that on January 17, 2019, the Prachand successfully engaged a moving aerial target with an air-to-air missile at the integrated test range at Chandipur, Odisha—the first ever air-to-air engagement in India by a helicopter. Its anti-tank capability is based on the Helina ATGM, which can hit targets between a range of 500 metres and seven kilometres.

LEAN MACHINE

The LCH Prachand **has the latest technology and has passed** rigorous flight tests



ANI

A SORTIE TO REMEMBER
Defence minister Rajnath Singh in the LCH in Jodhpur

- 5.8-tonne combat helicopter, capable of advanced self-protection,including stealth capability and crashworthy landing gear
- Armed with a chin-mounted 20mm turret twin-barrel gun, cluster bombs and 68mm rocket pods
- Maximum speed of 275kmph (148kt), combat radius of 500 kilometres
- Capable of high-altitude warfare, with an operational ceiling of 16,000 to 18,000 feet (5,490 metres)
- 45 per cent indigenous content
- IAF will have 65 LCH; the army will have 95 of the helicopters

THE LCH WAS PRESSED INTO SERVICE FOR **THE FACE-OFF WITH CHINA IN LADAKH** EVEN BEFORE THE APACHE WAS DEPLOYED

The LCH project has had its share of problems. Gp Capt. Nair points out that unlike the US or Europe, India does not have a large wind tunnel testing infrastructure that simulates flight conditions, so it took longer to develop the LCH. He adds that in 2010, when the first prototype came out, it had serious design and functional issues. It forced designers to go back to the drawing board for a major restructuring. IAF operates Russian-made Mi-25 and Mi-35 attack helicopters, along with the newly inducted Boeing-made Apache helicopters. While the Mi-25 and Mi-35 are set to be phased out, the IAF has a fleet of 22 AH-64E Apaches. Some experts say Apaches are not suited for high-altitude terrain and that they are designed for operations in West Asia. The force needed more attack helicopters, and the timing of Prachand's arrival, when copters are on the defence ministry's 'negative' list of imports, is perfect. "LCH overtakes Apache or any other combat chopper in the world," says Nair, reiterating that it is one of its kind.

HAL officials said that the flight testing on the four prototypes was carried out at various altitudes—from the sea level to the Siachen range as well as over extremely hot weather conditions over deserts. Further, integration of mission sensors such as electro-optical systems, helmet-mounted display systems, solid state data and video recorders are done and weapon firing trials completed. "A total of 2029 flights have been carried out on the prototypes till date," says an official.

Over 250 vendors are involved in the manufacturing of components, assemblies, fixtures, tools, test equipment and preparation of technical documentation, apart from 70 vendors involved in the indigenisation of various parts. Some believe that the helicopter has huge export potential and HAL has already obtained a no objection certificate from the MoD for its export to Malaysia, Thailand, Vietnam, Egypt, Indonesia, Ecuador and Nigeria.

Later this month, the army is going to deploy four Prachands in Assam's Missamari, within striking range of the forward areas along the Line of Actual Control in Arunachal Pradesh. It plans to have at least one squadron of LCH in each of its 17 corps.

"From air to ground and air-to-air capability, the LCH is able to operate on heights better than any helicopter in the world," says Gp Capt. Narang, talking it up. While operating in low altitude areas or plains, the Prachand can be fitted with counter-drone technology, since both China and Pakistan have significant drone capabilities.

Clearly, the LCH has been gathering fulsome praise from all quarters. Now it has to live up to the good press. ■